

Viyog: Separating Adversarial and Out-of-Distribution

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Abstract—Safety-critical systems must respond differently to two kinds of unusual input: out-of-distribution (OOD) inputs, which the system should abstain on, and adversarial (ADV) attacks, which it should reject outright. Most detectors give only one score and cannot tell these apart. We present Viyog, a plug-in add-on that needs no retraining. It separates OOD from ADV using one signal: how rough the quietest channels of the first convolutional layer become, read directly from the forward pass the model already computes. It needs only a tiny $O(C)$ calibration profile (about 0.3KB for 64 channels), versus 4.5–40 MB for feature-distance baselines. Viyog detects adversarials at held-out AUROC 0.980, versus 0.659 for the best confidence-based score we test. Paired with a logit score, it gives the first practical three-way ID/OOD/ADV separator, reaching balanced accuracy 0.798 on our 17-architecture panel and 0.863 on GTSRB. Across three datasets and four attacks, Viyog adds no parameters and costs only 2.28% of a model’s compute and 3.9% of its latency. Code: pypi.org/project/viyog. Live demo: huggingface.co/spaces/amanyagami/viyog.

Index Terms—Out-of-distribution detection, adversarial robustness, feature norms, deep neural networks, safety-critical systems, embedded AI, architecture generalization.

I. Introduction

Deep neural networks (DNNs) now achieve state-of-the-art performance in safety-critical domains: autonomous driving, medical imaging, embedded vision. A deployed system must also detect abnormal inputs and respond appropriately before acting on a prediction, which standard classification pipelines do not do [1].

Two kinds of anomalous input require opposite responses (Fig. 1). OOD inputs fall outside the training distribution, so the right response is to abstain. Adversarial (ADV) inputs are crafted to force a misclassification, so the right response is to reject them outright. Treating the two the same way is costly either way: it wastes a defence on a benign input, or it lets an attack through silently. Most lightweight baselines evaluated here do not draw this distinction at all. We investigate a signal that does: one found in the network’s first layer.

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We identify a consistent pattern in the first layer’s activations that holds across every model we test. Some first-layer filters barely respond to normal images at all; we call these dormant channels. Under the digital attacks we evaluate, these otherwise-quiet channels turn rough and noisy: the attack’s perturbation spreads energy across many spatial frequencies at once (a broadband signal), unlike the smooth, low-frequency structure of natural images. Clean (ID) and out-of-distribution (OOD) images leave these channels smooth. Prior work also reports early-layer and frequency-domain effects of adversarial perturbations [2], [3]. This is a pattern we observe under our threat model, not a guarantee that holds for every possible attack.

Before arriving at the method in this paper, our first attempt to exploit this pattern simply thresholded the raw magnitude of first-layer activations (we call this earlier, simpler attempt the L_∞ -form): adversarial inputs tended to fall below the typical in-distribution value, OOD inputs at or above it. This worked well when the OOD input was very different from the training data (far-OOD, e.g. a different dataset entirely), but on OOD inputs that only looked slightly different from the training data (near-OOD), it performed barely better than random guessing. The method we present instead measures the shape of the roughness in dormant channels, not its raw size: a scale-normalised total-variation (TV) statistic (Viyog, Sec. IV). Scaling every activation up or down by the same factor leaves this statistic unchanged (it is 0-homogeneous); this does not cover every perturbation that preserves overall magnitude, some of which can still change spatial roughness. Viyog is complementary to logit OOD scores and yields three-way ID/OOD/ADV separation (Sec. VI); the pattern holds across CNN and transformer architectures and across 32×32 and 224×224 inputs. We ground this in a formal Proposition (Section III) that gives a deterministic bound for affine pre-activations under explicit smoothness, residue-roughness, and residue-size conditions.

This paper makes five contributions:

- C1: The Viyog scoring statistic. A gradient-free statistic computed from the first layer’s dormant (quietest) channels that separates ADV from OOD where the older, magnitude-based approach cannot (mechanism confirmed on real ResNet-50 features, Fig. 2b). $V(x)$ is a continuous score; we only convert it to a yes/no decision at the final deployment step. The raw L_∞ score is kept throughout as a baseline for comparison.
- C2: Theory for the shape statistic. A formal proof (Sec. III-B) that our statistic does not change when activations are scaled uniformly, plus a bound on how

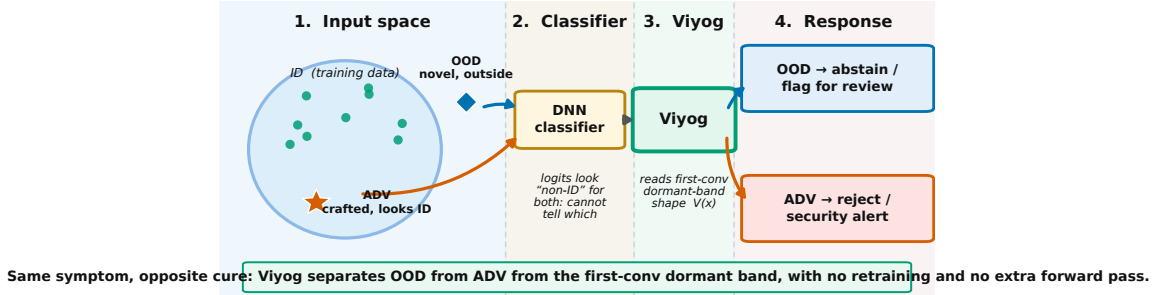


Fig. 1: The core problem: OOD and ADV inputs require opposite responses, yet both fool the classifier. (1) Input space: an OOD (out-of-distribution) input lies outside the training cluster; an adversarial (ADV) input is deliberately crafted to sit inside the cluster while forcing a wrong prediction. (2) Classifier: both can produce logits that the evaluated output-side detectors do not reliably distinguish. (3) Viyog: reads the dormant-band roughness $V(x)$ of the first convolutional activation map (a gradient-free statistic with $O(C)$ stored state) and separates the two. (4) Response: OOD → abstain / flag for review; ADV → reject / security alert. Conflating them wastes adversarial defences on benign inputs or silently accepts attacks.

much it can change under a real adversarial perturbation $e_c = w_c * \delta$. This guarantee holds only under stated smoothness conditions on the pre-nonlinearity layer; we test the nonlinear (ReLU/GELU) case empirically, so the observed ID/OOD/ADV ordering is explained by theory under these conditions, not guaranteed for every attack. A more general bound covering the older L_∞ baseline is in the supplementary material.

C3: Where the old approach fails, and how Viyog fixes it. We show that the older, magnitude-based L_∞ approach is weak specifically on OOD inputs that closely resemble the training data, and that Viyog recovers performance there. Combined with a logit-based OOD score in a lightweight classifier, the full detector panel reaches balanced accuracy 0.798 (CIFAR-100, 17 archs) / 0.863 (GTSRB, 6 archs); the logit+Viyog pair alone reaches 0.758/0.768. This adds only $O(C)$ state (about 0.3KB at 64 channels) and holds for ConvNeXtV2, ViT-B/16, and Swin-T at 224×224 .

C4: Broad empirical testing. We test on the datasets in Table II under four L_∞ attacks. Under a simpler, single-direction evaluation, the raw L_∞ score beats ten established baselines by up to +24.9 AUROC points. A stricter, directionless re-evaluation over the full 17-architecture panel, which disqualifies any score whose high/low meaning flips between models, is reported in Table III.

C5: Adaptive attack analysis. An attack that targets only the magnitude-based score is caught every time (0% evasion); an attacker aware of both signals succeeds only 8% more often. A defence that randomly changes which dormant channels are read, so an attacker cannot target a fixed set, restores detection to 0.82 (Fig. 12).

II. Background and Related Work

A. OOD Detection Methods

Table I categorises representative prior work; surveys are in [4], [5]. Most evaluated detectors assign a single anomaly score rather than an explicit OOD-versus-ADV

TABLE I: Representative anomaly detectors and their OOD-ADV separation capability in our evaluation. † Viyog scores both T2 and T3 above chance while remaining gradient-free and state-minimal.

Family	Examples	Score	Sep. OOD/ADV?	Cost
Confidence	MSP, Energy, ODIN	logit	✗ both high conf.	$O(1)$
Distance	Maha, KNN, ViM	feat. dist.	✗ ADV inside manifold	$O(D^2) - O(ND)$
Uncertainty	MCD	post. var.	✗ single score	30× fwd
Joint	UNICAD [7]	proto./DAE	partial	trained modules
Viyog†	this work	shape V	✓ T20.980 / T30.844	$O(C)$

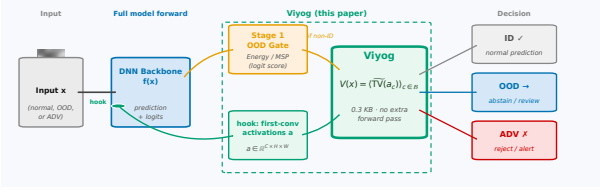
decision. Szyk et al. [6] show that rankings are protocol-sensitive, motivating our multi-seed evaluation (Sec. V).

B. Adversarial Attacks, Defences, and Early-Layer Effects

Adversarial attacks [8] solve $\min_{\delta} \|\delta\|_p$ s.t. $f(x+\delta, \theta) \neq y$. Common attacks include FGSM [9], PGD [10] (both L_∞), and L_2 -constrained CW [11] and DeepFool [12]. A fundamental tension exists between robustness and accuracy [13], [14].

Defences fall into several families: input preprocessing [15], feature denoising [16], generative purification [17]–[20], and adversarial training [10], [14]; ensemble and Jacobian-based methods add robustness at extra inference cost. None of these target the OOD/ADV disambiguation problem: they assume the threat class is known in advance, precisely the assumption Viyog relaxes.

Early-layer effects are well documented: the first convolutional layer is disproportionately critical for robustness [3] and acts as an implicit noise filter [2] (the front-neuron concept formalised here); pruning low-activation neurons improves robustness by forcing reliance on high-magnitude responses [21], and restoring suppressed activations [22] presupposes the suppression Viyog detects. Filter-statistic [23] and frequency-domain [24], [25] analyses likewise distinguish adversarial perturbations from natural shift at early layers, but none exploits the OOD/ADV asymmetry, which motivates the conditional pre-activation proposition of Section III.



(a) Candidate Viyog two-stage pipeline. A normal forward pass drives a first-stage OOD gate (logit-based scores such as Energy [29] or MSP [30]) and exposes the first-conv activation map $a = f_0(x)$ through a hook. Flagged inputs enter Viyog, which computes $V(x)$, the dormant-band roughness of a , and routes to ID (pass), OOD (abstain / flag), or ADV (reject / alert), at $O(C)$ state and no extra forward pass.

C. Joint OOD and Adversarial Detection

A small body of work attempts OOD–ADV separation directly, but at a cost Viyog avoids. Zhang et al. [26] propose an OOD detector via adversarial gradient-attribution paths, which needs gradient computation at inference and is not an OOD-vs-ADV router. Feng et al. [27] propose a unified open-world perspective but require joint training with anomaly labels. Walkowiak et al. [28] combine outlieriness scores and feature extraction; UNICAD [7] unifies attack detection and novel-class identification. Feature-statistic [23] and disentangled-representation methods have also been proposed. Unlike these gradient-attribution or joint-training approaches, Viyog runs in a single gradient-free forward pass as a plug-in second stage atop any existing non-ID detector, with no architectural changes. The reported protocol uses a small labelled ID/OOD/ADV calibration split to fix score polarity and operating points (the optional three-way head uses the same split, Sec. VI); this efficiency advantage is quantified next.

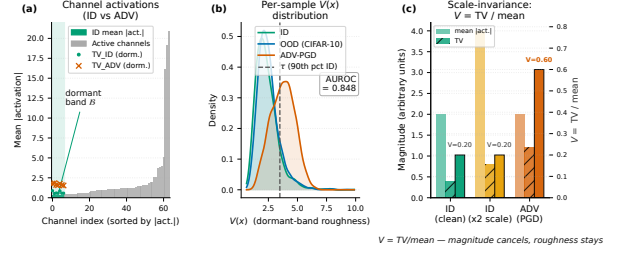
D. System Efficiency and Deployment Motivation

For embedded systems inference cost is a primary constraint [1], yet prior detectors are expensive: ODIN needs a gradient-norm backward pass, Mahalanobis a full covariance, MCD 30 forward passes ($\sim 30\times$ a single inference), and purification a diffusion pass. Viyog adds zero parameters and no gradient computation, reading a spatial reduction off a first-layer activation hook instead; the pipeline below motivates a candidate system integration.

The Viyog two-stage pipeline (Fig. 2a) partitions inputs flagged by a first-stage non-ID detector into OOD and ADV; its errors and end-to-end recall remain relevant in deployment (Sec. IX).

III. Theoretical Motivation and Analysis

We formalise a sufficient condition under which a rough additive residue raises the dormant-band statistic at an affine pre-activation (Prop. 1). The result proves uniform-rescaling invariance in the zero-offset limit and conditional



(b) The Viyog mechanism confirmed on real ResNet-50 features (CIFAR-100, $n=2000$ samples per class). (a) Dormant-band channels (bottom 10% alive, green) have the smallest denominators, amplifying injected roughness. (b) ID and OOD co-cluster at low $V(x)$; ADV-PGD spikes high. (c) Uniform-rescaling check (P1): doubling activation magnitude leaves V unchanged at the chosen ϵ ; the adversarial example has the elevated roughness ratio here.

separation; it is not a worst-case guarantee for arbitrary attacks or nonlinear post-activations. It rests on the convolutional structure shared by every architecture we test: strided convolutions in ResNet [31], DenseNet [32], and ConvNeXtV2 [33]; and patch-embedding projections (mathematically equivalent to strided convolutions) in ViT [34] and Swin-T [35]. Prior frequency-domain work supports the idea that such perturbations concentrate at early layers [15], [24], [25].

A. Why the First Layer?

Natural-image smoothness (empirical premise A1). Natural image power spectra decay as $P(f) \propto f^{-2}$: low-frequency components (slow gradients, textures) dominate; rapid pixel-to-pixel oscillations are negligible. Each first-layer channel c produces a feature map $a_c \in \mathbb{R}^{H \times W}$, the response of filter w_c to input x . For the tested natural inputs, a_c is typically smooth: its Total Variation $\text{TV}(a_c)$ (mean pixel-to-pixel jump, defined formally in Eq. (1)) is small relative to the channel’s mean magnitude $|a_c|$. Proposition 1 makes the required per-channel inequality explicit; natural OOD inputs that violate it are a failure mode rather than an exception to the premise.

Rough adversarial residue (empirical premise A2). Gradient attacks add a perturbation δ to raise the loss. At an affine first layer, each filter w_c maps it to the exact additive residue $e_c = w_c * \delta$. The four evaluated L_∞ attacks usually produce high relative TV in the selected channels. An L_∞ constraint alone does not impose this spectral property, however; a smooth or detector-aware perturbation can violate A2.

Why a normalised ratio, not a raw norm (0-homogeneity). A raw L_∞ norm measures magnitude, not shape. OOD images can produce larger or smaller first-layer activations than ID images, so $\|a_c\|_\infty$ is direction-inconsistent; an attacker can further append a norm-preserving penalty to keep $\|a'_c\|_\infty \approx \|a_c\|_\infty$ while misclassifying. The normalised ratio $\text{TV}(a_c) = \text{TV}(a_c) / (|a_c| + \epsilon)$ (small $\epsilon > 0$ for stability) addresses magnitude variation: TV and $|\cdot|$ are both 1-homogeneous, so their ratio is 0-homogeneous when

the stabilising offset is zero (P1 below). Thus uniform feature-map rescaling does not change the idealised ratio. Preserving a norm while changing spatial structure is not uniform rescaling and can change TV; the adaptive experiments measure, rather than rule out, that case.

B. The Dormant-Band Roughness Statistic

The statistic. Let $a = f_0(x) = W * x + b \in \mathbb{R}^{C \times H \times W}$ be the first-layer activation map (C channels, spatial size $H \times W$), with $a_c \in \mathbb{R}^{H \times W}$ the feature map of channel c , filter w_c , and bias b_c . Define:

$$V(x) = \frac{1}{|\mathcal{B}|} \sum_{c \in \mathcal{B}} \widetilde{\text{TV}}(a_c), \quad \widetilde{\text{TV}}(a_c) = \frac{\text{TV}(a_c)}{|a_c| + \epsilon}, \quad (1)$$

$$\text{TV}(a_c) = \frac{1}{2} (|\Delta_h a_c| + |\Delta_w a_c|),$$

where $\Delta_h a_c[i, j] = a_c[i+1, j] - a_c[i, j]$ and $\Delta_w a_c[i, j] = a_c[i, j+1] - a_c[i, j]$ are pixel differences, $\bar{u} = \frac{1}{HW} \sum_{i,j} u_{ij}$ is the spatial mean, and $\epsilon = 10^{-6}$ prevents division by zero. The dormant band is $\mathcal{B} = \text{bottom-}p\% \text{ of alive channels}$, where $\mathcal{A} = \{c : \mathbb{E}_{\text{ID}}[|a_c|] > \kappa\}$ ($\kappa = 10^{-4}$) excludes permanently silent channels.

For the analysis below, a denotes the affine pre-activation. A small denominator can make the dormant band sensitive to roughness, but the normalisation also makes dead-channel exclusion essential. Implementations that hook a post-nonlinearity are evaluated empirically; they are not covered by the affine proposition. Fig. 2b confirms this on real ResNet-50 features: $V(x)$ separates by type of signal (natural vs. attack), not by size of activation.

Proposition 1 (Conditional relative-roughness bound for an affine channel). Fix a channel $c \in \mathcal{B}$ of the affine pre-activation and write $a_c = w_c * x + b_c$, $\nu_c := |a_c| > 0$. For $x' = x + \delta$, affineness gives $a'_c = a_c + e_c$ exactly, where $e_c = w_c * \delta$. Define $r_c := |e_c|/\nu_c$, $\rho_c := \text{TV}(e_c)/|e_c|$ when $|e_c| > 0$, and $\eta_c := \epsilon/\nu_c$. Assume:

- (A1) Natural smoothness: $\text{TV}(a_c) \leq \beta_c \nu_c$; the same β_c upper-bounds the natural ID/OOD comparison class.
- (A2) Rough residue: $\rho_c > 0$ for the perturbation under consideration.
- (A3) Nonzero residue: $r_c > 0$.

Then:

- (P1) Uniform-rescaling identity: for $\lambda_c > 0$,

$$\widetilde{\text{TV}}(\lambda_c a_c) = \frac{\text{TV}(a_c)}{\nu_c + \epsilon/\lambda_c}.$$

Thus the statistic is exactly scale-invariant at $\epsilon = 0$, and only approximately so when $\epsilon/(\lambda_c \nu_c)$ is negligible.

- (P2) Deterministic two-sided bound:

$$\frac{[\rho_c r_c - \beta_c]_+}{1 + r_c + \eta_c} \leq \widetilde{\text{TV}}(a'_c) \leq \frac{\beta_c + \rho_c r_c}{|1 - r_c| + \eta_c},$$

where $[z]_+ = \max(z, 0)$. The upper bound becomes loose near $r_c = 1$, where cancellation can make the normalising denominator small.

- (P3) Conditional separation: if

$$\rho_c r_c > \beta_c(2 + r_c + \eta_c)$$

for every $c \in \mathcal{B}$, then the perturbed pre-activation has larger V than any natural comparison satisfying

- A1. The per-channel margin is at least

$$\frac{\rho_c r_c - \beta_c(2 + r_c + \eta_c)}{1 + r_c + \eta_c} > 0.$$

P3 is a sufficient condition, not a guarantee that A1–A3 hold for an unseen natural distribution or attack. ReLU is 1-Lipschitz, but that fact alone does not preserve a bound on a normalised ratio; GELU is neither globally monotone nor 1-Lipschitz. Consequently P2–P3 are not asserted for nonlinear post-activation maps. The supplementary material instead bounds the local GELU Taylor remainder, and Section VI reports the measured behavior.

Proof sketch (full proof in the supplementary material).

P1. Substitute the absolute 1-homogeneity of TV and $|\cdot|$ and divide numerator and denominator by λ_c .

P2. TV and the spatial L_1 norm satisfy triangle and reverse-triangle inequalities:

$$\text{TV}(a_c + e_c) \leq \text{TV}(a_c) + \text{TV}(e_c),$$

$$\text{TV}(a_c + e_c) \geq \text{TV}(e_c) - \text{TV}(a_c).$$

Also, $\nu_c |1 - r_c| \leq |a_c + e_c| \leq \nu_c(1 + r_c)$. Substituting $\text{TV}(e_c) = \rho_c r_c \nu_c$, adding ϵ to the denominator, and taking the nonnegative part of the lower numerator gives P2.

P3. Any natural comparison satisfying A1 has normalised TV at most β_c . Subtracting this from P2's lower bound gives the displayed margin; averaging positive per-channel margins over the fixed band gives the result. $\square$

In plain words. When a natural feature map is smooth and the additive affine residue is sufficiently large and rough, the bound predicts a higher V . The premise can fail for naturally rough OOD inputs or attacks that optimise the spatial statistic, so the prediction must be checked empirically under the intended threat model.

IV. The Viyog Method

A. Step-by-Step Derivation and Motivation

The first design choice, reading at the first convolutional layer, follows directly from the theory of Section III; the remaining three (restricting to the dormant band, using a total-variation shape statistic, and held-out calibration) are developed in the subsections that follow:

First convolutional layer. The first convolutional layer has an exact affine perturbation relation before its nonlinearity [2], [3], and, crucially, the adversarial residue has not yet been transformed or pooled. The shape statistic V peaks at the first conv (mean T2 0.965) and decays with depth in the six-model, two-attack depth sweep; it is therefore the lowest-cost and highest-AUROC layer in that experiment (Fig. 9).

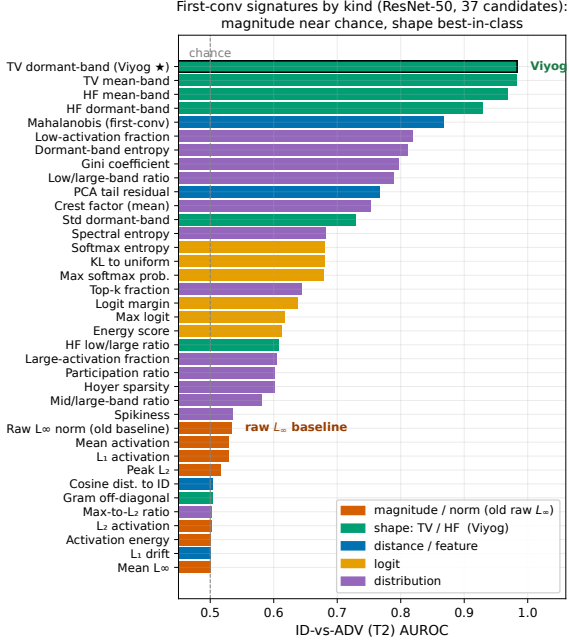


Fig. 3: First-conv signature ranking by ID-vs-ADV (T2) AUROC on ResNet-50 (37 candidates), coloured by family. Magnitude/norm (orange, including the raw L_∞ baseline) sits near chance (0.50–0.53); shape (green, TV/HF dormant-band) is best-in-class (≈ 0.98) with Viyog first. Distance, logit, and distributional statistics fall in between: shape, not magnitude, separates ADV from natural inputs here.

B. Decision Rule and Downstream Routing

Given a pool of non-ID inputs flagged by a first-stage detector, Viyog applies a second-stage partition: if $\text{Viyog}(x) > \tau$, ADV (apply adversarial defence, rejection, or security alert); if $\text{Viyog}(x) \leq \tau$, OOD (apply abstention, fallback, or incremental-learning trigger). A magnitude threshold allows precision–recall trade-offs. In the reported protocol, score polarity and threshold τ are fixed from a labelled ID/OOD/ADV calibration split and then applied once to disjoint test data (Alg. 1); the end-to-end gate uses the same held-out protocol (Sec. IX-A).

C. Dormant-Band Shape Statistic (Viyog)

The primary statistic $V(x) = \text{Viyog}(x)$ uses the dormant band of Eq. (1). Restricting it to active channels is essential: up to 25/64 first-conv channels are permanently silent (e.g. DenseNet-121) and would otherwise force a degenerate score. Across the 37-statistic battery V tops both the ID-vs-ADV and OOD-vs-ADV rankings (Fig. 3).

D. Algorithm and Complexity

Fig. 4 provides a step-by-step visual walkthrough of the full Viyog pipeline with activation-map images at each stage.

Algorithm 1 summarises the Viyog procedure: held-out calibration computes the per-channel ID profile, dormant band $\mathcal{B}$, polarity, and operating point; inference then

Algorithm 1 Viyog: Dormant-band shape statistic for OOD-vs-ADV separation

Require: Model f with first affine layer f_0 ; labelled calibration sets $\mathcal{D}_{\text{ID}}, \mathcal{D}_{\text{OOD}}, \mathcal{D}_{\text{ADV}}$; band fraction $\rho=0.1$; test sample x

- 1: Calibration (once per model):
 - 2: $p_c \leftarrow \frac{1}{|\mathcal{D}_{\text{ID}}|} \sum_{x \in \mathcal{D}_{\text{ID}}} |f_0(x)_c|$ $\triangleright$ per-channel ID mean activation
 - 3: $\mathcal{A} \leftarrow \{c : p_c > 10^{-4}\}$ $\triangleright$ drop permanently-dead filters
 - 4: $\mathcal{B} \leftarrow$ bottom- ρ of $\mathcal{A}$ by p_c $\triangleright$ dormant band
 - 5: $(d, \tau) \leftarrow$ fit polarity $d \in \{-1, +1\}$ and operating point on labelled calibration scores $\triangleright$ then freeze
 - 6: Inference (per sample, single forward pass):
 - 7: $a \leftarrow f_0(x)$ $\triangleright$ hook; no extra pass, no gradients
 - 8: $s \leftarrow \frac{1}{|\mathcal{B}|} \sum_{c \in \mathcal{B}} \widehat{\text{TV}}(a_c)$ $\triangleright$ dorm-band roughness, Eq. (1)
 - 9: if $d s > \tau$ then
 - 10: return ADV (reject / defend) $\triangleright$ high-frequency dorm-band residue
 - 11: else
 - 12: return OOD (abstain / incremental learn)
 - 13: end if
- Optional 3-way classifier: fit a Linear Discriminant Analysis (LDA) on the labelled calibration split over the feature vector [logit score, $V(x), \dots$]. This adds <0.1 KB of state and returns a three-class ID/OOD/ADV decision (Sec. VI).

adds only $\mathcal{O}(|\mathcal{B}| h_1 w_1)$ operations and $\mathcal{O}(C)$ stored state ($\mathcal{O}(C)$; about 0.3KB at 64 channels), measured as 2.28% of CNN MACs and 3.9% of H200 inference latency, with four to five orders of magnitude less stored state than the evaluated feature-distance detectors. We now validate it against eleven baselines across twenty model–dataset configurations.

V. Experimental Setup

A. Datasets and Models

All experiments use two primary ID datasets: CIFAR-100 [36] (224×224 , the main benchmark) and the safety-critical GTSRB [37] (224×224). Per-dataset OOD sets are drawn from SVHN [38], iSUN, LSUN, and Places365; MNIST serves as a supplementary near-OOD probe for DenseNet.

The model panel comprises 17 architectures spanning every major design family: ResNet-50/101/152 [31], DenseNet-121/161/169/201 [32], ConvNeXtV2 [33], ViT-B/16 [34], Swin-T [35], MobileNetV3-L, MobileNetV4-M, EfficientNet-Lite0, EfficientNetV2-L, EfficientViT-B1, FastViT-SA12, and EdgeNeXt-Small, all fine-tuned from ImageNet pretraining in PyTorch [39] without adversarial augmentation. (ResNet-18/34 and MobileOne-S1 were excluded: their archived first-layer features are identically zero, Table V.) Training-seed confidence intervals (≤ 3 seeds per architecture) are reported throughout; per-architecture std on T2 is ≤ 0.004 (Fig. 7).

How Viyog works: one forward pass, no extra parameters

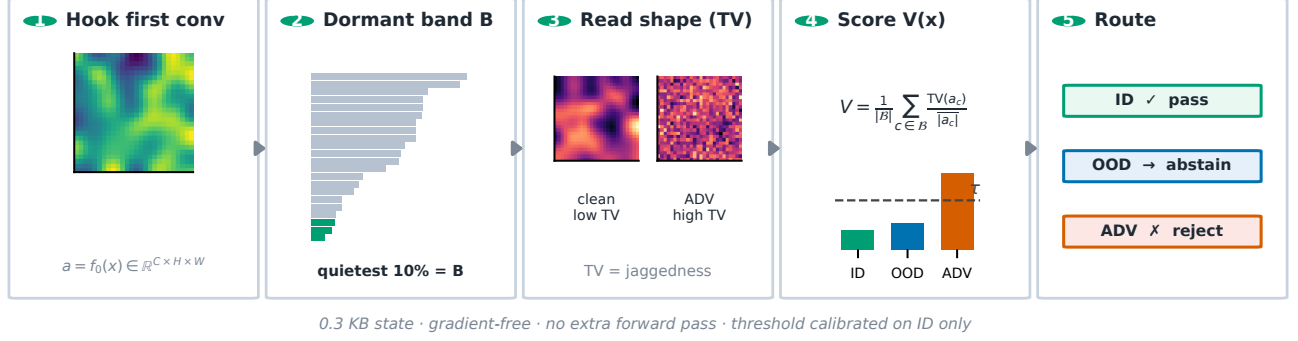


Fig. 4: How Viyog works, in five steps (one forward pass, no extra parameters). (1) Hook the first-conv activation map $a = f_0(x)$. (2) Rank channels by mean ID activation; the quietest 10% form the dormant band $\mathcal{B}$ (green). (3) Read spatial roughness (total variation) on $\mathcal{B}$: clean channels are smooth (low TV), adversarial channels are jagged (high TV). (4) Score $V(x)$, the mean normalized TV over $\mathcal{B}$: the calibration split fixes polarity and threshold τ . (5) Route the input: ID passes, OOD abstains, ADV is rejected. Full method and routing rule in Sec. IV (Alg. 1).

B. Attacks, Baselines, and Metrics

Four L_∞ attacks span the threat model: FGSM [9], BIM [40], PGD [10], and APGD-CE [41] via TorchAttacks [42] at $\varepsilon=8/255$. Eleven pytorch-ood [43] baselines, MSP, Entropy, MaxLogit, Energy, GEN, ODIN ($T=1000$), MCD (30 passes), Mahalanobis, KNN, OpenMax, and KL-Matching, are compared on AUROC, FPR@95TPR (false-positive rate at 95% true-positive rate), and AUTC (area under the threshold curve, critical for deployment where post-hoc tuning is infeasible). We evaluate three tasks: T1 ID-vs-ODD, T2 ID-vs-ADV (the security task), and T3 OOD-vs-ADV (the second-stage routing task).

Held-out calibration protocol. For the main CIFAR-100 evaluation, each pool is partitioned deterministically into 20% calibration and 80% test data (split seed 0). Calibration data alone determine the alive/dormant channels (absolute threshold $p_c > 10^{-4}$), KL-Matching references, score polarity, and operating points, which are then frozen and evaluated once on the disjoint test split, not a test-set oracle or $\max(\text{AUROC}, 1 - \text{AUROC})$. A $\times$ in Table III indicates that the raw score’s high/low direction varies across architectures; it is a diagnostic rather than a post-hoc test orientation rule [6]. Table II studies marked $\dagger$ predate the exclusion above.

Experimental coverage. Table II lists all experiments; the main sweep uses the held-out protocol above, and specialised studies use the protocol in their own section (depth sweep: Fig. 9; cost benchmark: Figs. 10–11; adaptive ladder: Sec. VII; seed CIs: Fig. 7; non-vision check: supplementary material). The adaptive ladder and training-seed CIs predate the MobileOne-S1/ResNet-18/ResNet-34 fix and have not been rerun on the corrected panel (Table V).

VI. Results and Discussion

A. Detection Quality Across 17 Architectures

With the evaluation protocol of Sec. V fixed, we first report raw detection quality across all 17 usable backbones.

TABLE II: Experimental coverage (see text for the calibration split and $\dagger$ notation).

Experiment	Coverage	Measures
Main T1/T2/T3	CIFAR-100, 17 usable archs, 20/80 held-out split	AUROC, FPR@95, recall
Measured cost	CIFAR-100, H200 (CUDA events)	latency, mem, energy
Layer-depth sweep	6 archs $\times$ 2 attacks $\times$ all depths	T2/T3 vs. depth
Adaptive A0–A3, EOT †	20 archs; norm/TV-aware + stochastic	evasion %, AUROC
Strong adaptive A4	6 archs (C100+GTSRB), held-out basis	detector floor
Training-seed CIs †	20 archs $\times$ ≤ 3 seeds	T2/T3 std
Non-vision sanity	1D signals, FGSM/PGD	T2 AUROC

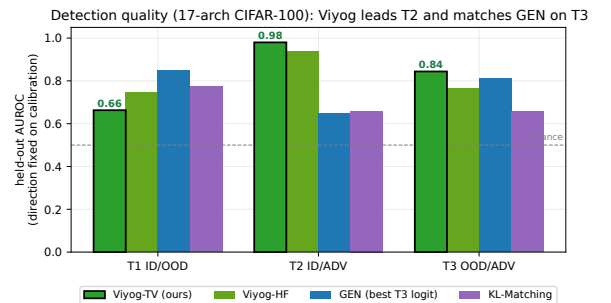


Fig. 5: Detection quality across the 17 CIFAR-100 architectures, by task and method family: Viyog leads T2/T3 (values in text below), while logit detectors win the T1 ID/ODD task (0.86). Tasks are scored directionlessly. At a strict 5%-FPR point mean ADV recall is 0.957 (median 0.964); Sec. IX-A reports an end-to-end cascade analysis.

Under the held-out 17-architecture protocol (Table III, Fig. 5), Viyog leads on the security task T2 (0.980) and T3 (0.844) with $O(C)$ state. The five evaluated confidence/logit scores (Energy, MSP, MaxLogit, Entropy,

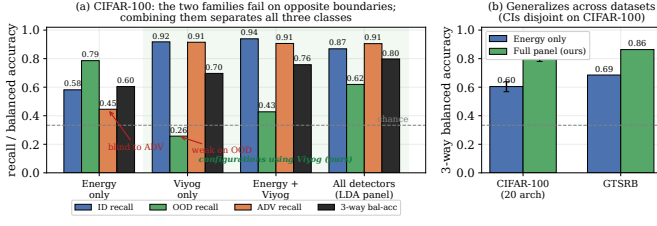


Fig. 6: Combining a logit detector with Viyog yields three-way ID/OOD/ADV separation: each covers the other’s blind spot. Energy alone catches low-confidence OOD but is blind to confident adversarials; Viyog catches adversarials but is weak on smooth OOD. (a) Per-class recall (CIFAR-100, 17 archs): Energy 0.445 on ADV, Viyog 0.257 on OOD; Energy + Viyog raises 3-way balanced accuracy to 0.758, full panel 0.798. (b) Transfers to GTSRB: full panel 0.863, logit+Viyog 0.768; bootstrap 95% CIs are disjoint ($[0.77, 0.83]$ vs. $[0.58, 0.65]$); no retraining or extra forward pass, at $O(C)$ state.

TABLE III: Held-out evaluation across the 17-architecture CIFAR-100 panel (of an original 20; MobileOne-S1, ResNet-18, and ResNet-34 excluded, Sec. V). Mean AUROC / FPR@95TPR (%); $\times$ = direction-inconsistent; best per column in bold; shaded row = primary detector. Five of six logit baselines are direction-inconsistent on T2; the signal is carried by first-conv shape, not magnitude or logits.

	AUROC / FPR@95TPR (%)				
Method	T1 ID/OOD	T2 ID/ADV	T3 OOD/ADV	State	
Logit baselines (pytorch-ood)					
Energy	0.863 / 52.1	0.600 $\times$ / 98.0	0.790 / 80.5		4 B
MSP	0.837 / 61.9	0.621 $\times$ / 98.4	0.796 / 79.1		4 B
MaxLogit	0.864 / 52.5	0.599 $\times$ / 98.6	0.794 / 79.5		4 B
Entropy	0.848 / 62.1	0.627 $\times$ / 98.7	0.803 / 77.0		4 B
GEN	0.848 / 63.4	0.647 $\times$ / 98.6	0.812 / 75.6		4 B
KL-Match	0.775 / 82.5	0.659 / 79.7	0.655 / 83.7		4 B
First-layer statistics (ours)					
FirstConv- L_∞ (baseline)	0.562 $\times$ / 98.1	0.515 $\times$ / 87.6	0.577 / 75.0		8 B
Viyog-HF (high-freq. dorm.)	0.745 / 83.3	0.939 / 25.9	0.763 / 59.4		$O(C)$
Viyog (primary)	0.663 / 96.5	0.980 / 4.9	0.844 / 37.4		$O(C)$

GEN) score at most 0.647 on T2; direction-consistent KL-Matching reaches 0.659. A three-way supervised head reaches 0.798 balanced accuracy (Fig. 6), and the T3 signal stays above chance across all 17 usable architectures, four attacks, and three datasets (Fig. 8; GTSRB is the second ID dataset).

The failure modes are complementary: logit scores collapse on T2 (blind to high-confidence adversarials), while feature-distance detectors (Mahalanobis, KNN) trail on T3, having pulled perturbations back toward the ID manifold at deep layers [44]. Among the compared single statistics, the first-conv shape statistic separates both (Sec. III-A).

A limited training-seed check (at most three seeds per architecture) shows small T2 variability in this panel (Fig. 7); broader training and data-split replication remains future work.

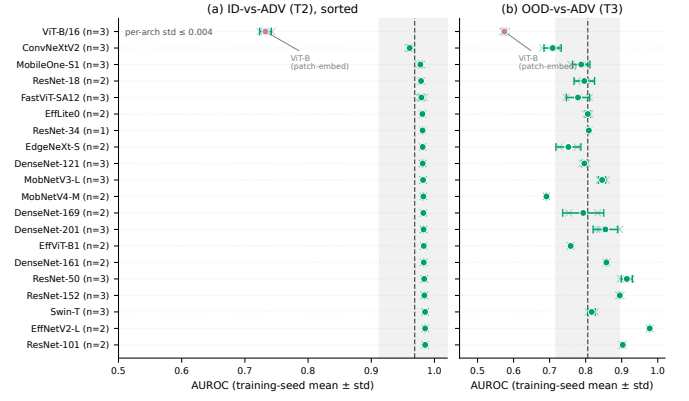


Fig. 7: Training-seed reproducibility across the full 20-architecture panel. Per-architecture mean $\pm$ std (circles), individual seeds (faint crosses), panel mean (vertical line), ± 1 overall std (band). (a) T2 seed std ≤ 0.004 across all 20 backbones, an order of magnitude below the cross-architecture spread (ViT-B the semantic-token outlier). (b) T3 similarly stable (std ≤ 0.06).

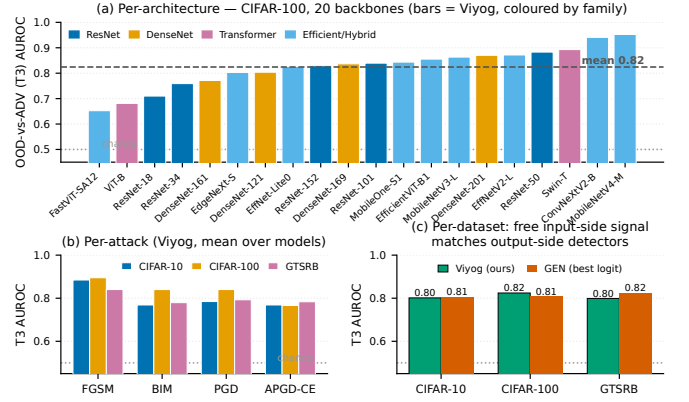


Fig. 8: The OOD-vs-ADV (T3) signal is consistent across architectures, attacks, and datasets. (a) Per-architecture T3 AUROC on 17 usable CIFAR-100 backbones (mean 0.844, range 0.68–0.95, every backbone above chance). (b) T3 across four attacks (FGSM, BIM, PGD, APGD-CE) and three datasets, weakest case ≥ 0.77 . (c) Viyog matches the strongest logit detector (GEN) within ± 0.02 at $O(C)$ state and no extra forward pass, yet stays complementary to it (Fig. 6).

B. Near-OOD Difficulty and Shape-Depth Sweep

We break the T3 result by OOD difficulty and compare the shape read across depth; ID/OOD channels are generally smoother than their adversarial counterparts (Fig. 14).

Harder near-OOD and texture splits. Near-OOD and texture-biased OOD inputs can themselves perturb early activations, so we split the 17-architecture T3 result by difficulty. Viyog reaches 0.96 on far-OOD and degrades gracefully but stays above chance on near-OOD (0.74 mean, up to 0.95) and texture-OOD (0.67); near-OOD is the harder regime. Combining the TV read with a high-frequency diagnostic recovers 0.92 on texture-OOD,

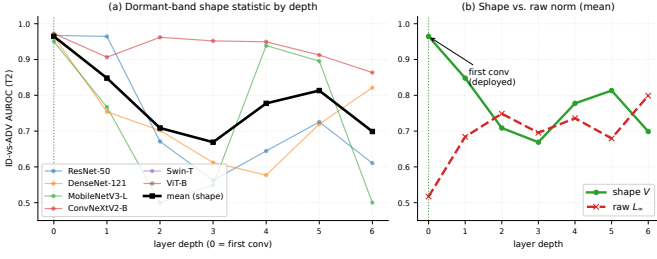


Fig. 9: The first conv is the strongest layer in the evaluated depth sweep. ID-vs-ADV (T2) AUROC of the dormant-band shape statistic V vs. the raw L_∞ statistic at every depth, averaged over six architectures (ResNet-50, DenseNet-121, MobileNetV3, ConvNeXtV2, Swin-T, ViT-B) and two attacks (PGD, APGD). The raw norm (red) is weakest at the first conv (0.52), peaking only deep (0.80, depth 6); shape (green) beats the best raw-norm layer at every depth.

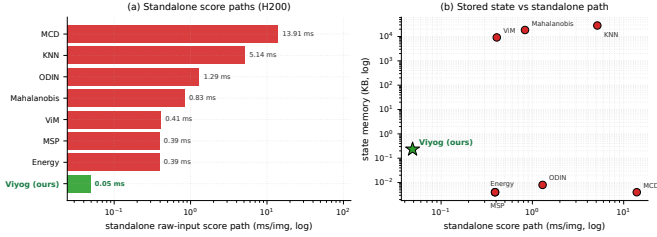


Fig. 10: Standalone score-path runtime and stored state (CIFAR-100; mean over ResNet-50, MobileNetV3, and DenseNet-121; H200 with CUDA synchronisation). (a) These raw-input timings traverse different portions of the backbone: Viyog stops after the first layer, whereas logit and deep-feature scores require later layers, so they describe standalone score paths, not deployed speedup. The Viyog path is 0.05 ms/img. (b) 0.23 KB for the plotted 64-channel profile against 9–28 MB for the distance detectors (10^4 – $10^5 \times$ smaller). Incremental overhead for a normal full inference is reported separately in Fig. 11.

though this is an oracle upper bound (it picks the better statistic per architecture using test performance), not a deployable rule (see supplementary material for the per-difficulty breakdown). In a six-architecture, two-attack depth sweep, the shape read peaks at the first layer (mean T2 0.965, T3 0.918) and decays with depth (Fig. 9).

C. Computational Efficiency

We next compare the accuracy and cost of the evaluated detectors.

Figs. 10–11 quantify the cost (values in Fig. 11): Viyog uses $O(C)$ state (about 0.3 KB at 64 channels) with no extra forward pass; edge-energy values below are roofline projections, not on-device measurements. The standalone raw-input score-path timings in Fig. 10 are not deployed speedup measurements. In complexity terms, logit scores and Viyog both add zero extra forward passes ($O(1)$ and $O(C)$ state, respectively, versus $O(D^2)$ for Mahalanobis/ViM and $O(ND)$ for KNN, each requiring one

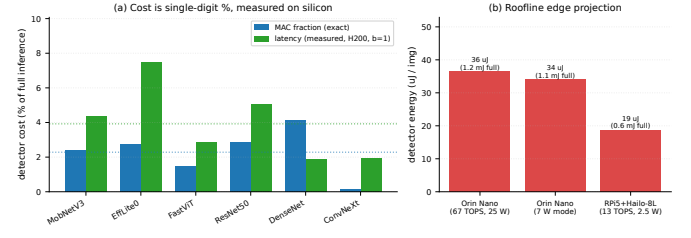


Fig. 11: Measured H200 detector cost and edge projection. (a) Per-architecture cost of the primary $V(x)$ detector as a fraction of full inference: the exact MAC fraction (mean 2.28%) and the measured batch-1 H200 latency (CUDA events; mean 3.9%) show a single-digit-% tax across CNNs, a ViT, and a ConvNeXt. (b) Roofline projection of per-image energy on edge accelerators (Jetson Orin Nano, RPi5 + Hailo-8L): 19–37 μ J versus 0.6–1.2 mJ full inference; these are optimistic lower bounds, and memory, runtime, and power effects need measurement on each target device.

extra forward pass); an optional “LDA head” — a tiny linear discriminant fit over the detector scores that turns the binary detector into a 3-class ID/OOD/ADV classifier ($K=3$) — adds only <0.1 KB more state and no extra forward pass.

Together the AUROC, AUTC, latency, and memory results place Viyog favorably among the evaluated methods.

VII. Adaptive Attack Evaluation

A rigorous adversarial evaluation must include an adaptive adversary who knows the detection mechanism and specifically targets it.

Threat model. The attacker aims at both misclassification and detector evasion under L_∞ , $\varepsilon=8/255$. Fig. 12 shows the full A0–A3 knowledge ladder; the defender’s hook, band $\mathcal{B}$, and threshold τ are fixed at deployment.

A. Adaptive Attack Formulation

A norm-aware attacker knowing that Viyog monitors first-layer L_∞ norms would optimise perturbations to preserve these norms while inducing misclassification. We maximise the following penalised PGD objective:

$$\mathcal{L}_{\text{adap}} = \mathcal{L}_{\text{CE}}(f(x+\delta), y) - \lambda (\|f_0(x+\delta)\|_\infty - \|f_0(x)\|_\infty)^2, \quad (2)$$

where $\lambda > 0$ controls the penalty strength; we sweep L_∞/L_2 /combined modes ($\varepsilon=8/255$, $\alpha=2/255$, 10 PGD steps). The reported behavior is specific to these attacks, budgets, steps, and penalty sweeps.

B. Results and Security Implications

The ladder of Fig. 12 measures what each rung of attacker knowledge costs, from black-box PGD (A0) through a both-aware adaptive adversary (A3) to a stochastic-band defence (EOT).

With the penalty active, $\|f_0(x_{\text{adap}})\|_\infty \approx \|f_0(x)\|_\infty$, keeping the raw score in-distribution, yet the dormant-band shape read still achieves 0% measured evasion under

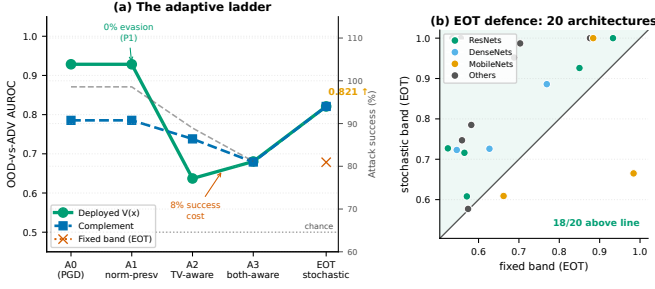


Fig. 12: Defence-in-depth against an escalating adaptive adversary (mean over 20 architectures). Attacker knowledge grows left-to-right: A0 (PGD), A1 norm-preserving, A2 TV-aware (white-box), A3 both-aware, EOT stochastic defence. (a) A1 achieves 0% evasion empirically; A2 erodes V to 0.64 while the complement holds at 0.74; EOT restores V to 0.82; A3 suppresses both only at an 8% attack-success cost. (b) 18/20 architectures are above the diagonal, supporting stochastic band-randomisation here, not a robustness certificate.

A1 (as above) — an observation that does not follow from P1, since holding a norm fixed is not uniform rescaling.

The result holds on a second dataset. On GTSRB (ResNet-50, ConvNeXtV2, ViT-B) at max penalty λ , the both-aware attacker keeps $\geq 80\%$ success, yet the combined $\max(\text{dorm}, \text{HF})$ detector never drops below AUROC 0.775: single-channel evasion is backstopped by the complement (ResNet-50 dorm $0.83 \rightarrow 0.51$ but HF stays 0.72). The attacker must either accept detection ≥ 0.775 or surrender most of its success, on both datasets.

Behavior of the evaluated attacks. Standard (A0) attacks [9]–[12], [41] naturally suppress first-layer norms and are often detected; the bespoke penalty of Eq. (2) targets that behavior directly. A different white-box optimiser may find another trade-off, so these results are not a universal cost of evasion [42].

Failure-mode caveat. Evading the global L_∞ score may route a sample toward ID or OOD. Abstention is safer only if the first-stage gate still flags that sample; otherwise the adversarial can remain undetected.

C. The Robustness–Evasion Trade-Off

In the tested optimisation, norm preservation and misclassification compete, costing the both-aware attacker 8% success, an empirical frontier for this optimiser, not a fundamental lower bound. A stronger held-out-basis (A4) attacker, re-optimising its perturbation basis directly against the statistic, shows a similar frontier ($\sim 15\%$ of attack success surrendered; see supplementary material).

VIII. Generalisation Beyond CNNs

The affine part of Proposition 1 applies to convolution and patch-projection pre-activations, but its empirical premises still need checking per model. We test three architecturally distinct models fine-tuned on CIFAR-100 at 224×224 from ImageNet-21k weights: ConvNeXtV2-Base [33] (depthwise convolution), ViT-B/16 [34] (pure

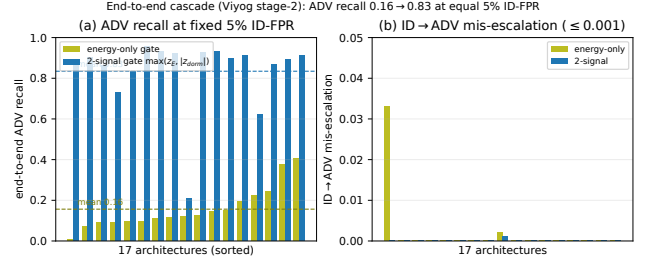


Fig. 13: Two-stage cascade over the 17-architecture panel at a disjoint-calibration 5% ID-FPR (values in Table IV). (a) Energy-only vs. two-signal gate ADV recall. (b) ID $\rightarrow$ ADV mis-escalation stays negligible.

self-attention), and Swin-T [35] (hierarchical shifted-window), each with an affine strided-convolution or patch-embedding first layer (4×4 , 16×16 , and 4×4 respectively), so the affine perturbation relation of Sec. III holds exactly at the pre-activation stage. Our implementation reads post-activation maps, for which Proposition 1 gives no guarantee (the GELU Taylor remainder is discussed in the supplementary material).

On all three, the dormant-band roughness $V(x)$ separates ADV from ID/OOD as in the 32×32 CNN results, and the separation is stable over the evaluated seeds and data resamples (Fig. 7); at most three training seeds per architecture do not establish seed invariance.

These results provide evidence that the signal transfers to several convolutional and patch-projection architectures; they do not prove universal architectural generality.

IX. Downstream Utility: Three-Way Classification

Viyog turns a binary anomaly detector into a three-way ID/OOD/ADV classifier, a distinction AUROC/FPR alone cannot capture. Without it, a system must apply one response to every non-ID input: wasting purification [18]–[20] or denoising [16] on benign OOD, or leaving attacks undetected. The same distinction matters for class-incremental learning [45], [46], where OOD inputs should become new classes but ADV inputs must not contaminate training.

A. End-to-End Cascade and False-Positive Propagation

We report a two-stage analysis (non-ID gate $\rightarrow$ Viyog) over the 17-architecture panel $\times$ 10 calibration/test split seeds at a matched 5% ID-FPR (Table IV, Fig. 13). Both stages’ thresholds are fit on each seed’s calibration split only and evaluated once on its disjoint test split (heldout_calibration_v1, Sec. V); this is the same protocol as the main results and excludes MobileOne-S1, ResNet-18, and ResNet-34.

Adding the first-conv deviation ($\max(z_{\text{energy}}, |z_{\text{dorm}}|)$, same 5% ID-FPR) to the Energy-only gate substantially raises ADV recall (Table IV); the weakest architecture is ViT-B at 0.29, since its patch embedding has no dormant band, and ID $\rightarrow$ ADV mis-escalation is zero on 15/17 architectures.

TABLE IV: Two-stage recall at a disjoint-calibration 5% ID false-positive rate (mean over 17 archs $\times$ 10 split seeds).

Stage-1 gate	ADV recall	OOD recall	ID $\rightarrow$ ADV
Energy only	0.157	0.343	0.002
Energy $\oplus$ dorm-z (2-signal)	0.834	0.424	0.000

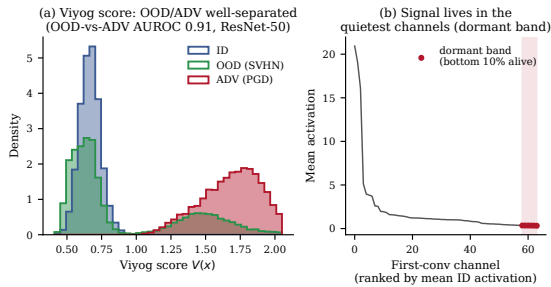


Fig. 14: The dormant-band signature. Per-image dormant-band TV profile for ID (green), OOD (blue), and adversarial (red) inputs on ResNet-50 (CIFAR-100, five OOD sets, PGD attack). ID and OOD share a low, smooth profile; adversarials inject high-frequency structure, separating at AUROC 0.91.

TABLE V: Current limitations and required follow-up.

Limitation	Current boundary and required follow-up
(i) Theory	Conditional affine pre-activation bound; nonlinear hooks are empirical, with no worst-case guarantee.
(ii) Threat	Four digital L_∞ attacks at 8/255 and finite adaptive sweeps; test other norms, physical attacks, and unseen objectives.
(iii) Accuracy	T3 FPR@95 is 37.4%; cascade OOD recall is 0.424 and ViT ADV recall is 0.29; these are not stand-alone safety evidence.
(iv) Generality	Three image datasets, ≤ 3 seeds/model, one synthetic 1D task. Excludes MobileOne-S1/ResNet-18/ResNet-34 (zeroed features, Sec. V); $\dagger$ -marked studies (Table II) predate this and still report 20.
(v) Hardware	H200 latency is measured; edge energy is projected and requires measurement on target devices.

B. Limitations and Future Work

Table V states the principal scientific limitations. In particular, the affine proposition is conditional and the adaptive evaluation is not a certificate.

X. Conclusion

We proposed Viyog, a gradient-free method that uses first-layer dormant-band roughness to separate OOD from ADV inputs. Under the evaluated digital L_∞ threat model, Viyog reaches T2 AUROC 0.980 across the 17-architecture held-out panel. Proposition 1 gives a sufficient pre-activation separation condition; it is not a guarantee for arbitrary attacks, natural shifts, or nonlinear hooks. In the adaptive suite, A1 has 0% measured evasion, while TV-aware attacks reduce performance.

Combined with logit scores, Viyog reaches three-way balanced accuracy 0.798 on CIFAR-100 and 0.863 on GTSRB, adding no learned parameters and $O(C)$ state at

2.28% CNN MAC / 3.9% H200 latency overhead. We do not yet claim Viyog is ready for stand-alone deployment: its false-positive rate on T3 is still fairly high, some cases remain hard, adaptive-attack coverage is limited, and projected energy cost on edge hardware is untested. Table V lists the follow-up work needed to close these gaps.

Artifact Availability

The evaluated MIT-licensed source, tests, paper source, and reproduction entry point are archived as release v0.1.3-codes26-ae-6 at Zenodo [47] and mirrored by the exact GitHub release. The detector is distributed as `viyog==0.1.3`, and the checkpoints are pinned at Hugging Face. The canonical entry point is `reproduce_t1.sh`; the precomputed adversarial dataset and live Space linked from the archive README are optional and are not required by that script.

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